Royce Barboz

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Selected Projects

LedgerQA | *Python, XBRL extraction, layout-aware parsing, agentic tool-calling, pgvector + Redis, FinanceBench* **GitHub**

- Answered financial questions over 84 10-K/10-Q/8-K filings from 32 companies ingested via SEC EDGAR, scoring 68% on FinanceBench's 150-question open-source set versus 24% for a text-RAG baseline; cited every answer to its exact `accession_number`, item label, and page while versioning facts with `valid_from`/`valid_to` to preserve restatements and 10-K/A amendments.
- Raised numeric accuracy from 50% to 72% by extracting figures from filers' XBRL tags and resolving custom extension tags through the XBRL calculation linkbase instead of reconstructing tables from HTML/PDF layout; eliminated LLM arithmetic errors by routing ratio and growth calculations through a code-interpreter sandbox, lifting computed-answer accuracy from 60% to 90%.
- Built document-type routing between XBRL and layout-aware engines with a messy-layout fallback that recovered 10 pre-2020 filings; capped agents at 6 tool calls with a 30s execution timeout, served Redis → pgvector → live EDGAR lookups at 12s p95 latency and $0.02/question across 40 full evaluation passes.

Verity | *LangGraph, LiteLLM, Chroma, AWS ECS Fargate, Pydantic, Datadog* **GitHub**

- Raised retrieval recall@5 from 0.72 to 0.84 and pipeline pass rate from 71% to 84% on a 31-ticket eval set by adding a Qwen3 reranker and fair-merge multi-query retrieval over ChromaDB.
- Grounded each customer-facing claim in a retrieved KB chunk via document-aware chunking and OpenAI embeddings over Chroma; enforced faithfulness and citation coverage as pass/fail gates in CI.
- Blocked prompt injection with layered regex and LLM detection plus a hard exit to human escalation; redacted PII before any NoSQL (DynamoDB) database or log write.
- Deployed on AWS ECS Fargate behind an ALB via Terraform, hardening security with Linux-based container configs and IAM least-privilege roles, with Datadog LLM Observability spans surfacing $0.0042 per ticket at 2.4 sec p95.

ProcureIQ | *LangGraph, FastAPI, GPT-4o, Azure AI Search, Terraform, Dynatrace, React/TypeScript* **GitHub**

- Built a four-agent LangGraph vendor risk pipeline in Python (Market Scout to Policy Librarian to Risk Synthesizer to Human Review) that retrieves policy clauses from an Azure AI Search vector index and produces citation-backed Approve/Escalate/Reject briefs with 1–10 risk scores, routing dual hard-fails to human review via GPT-4o structured output; validated across 8 vendor scenarios.
- Eliminated fabricated risk scores across 12 partial-data test cases by engineering a graceful-degradation matrix: single Tavily retry with ERP-only fallback, blocking-vs-advisory policy classification, and confidence that decays algorithmically from missing sources, hard-capping incomplete assessments at 6 or below.
- Streamed live node-completion events to a React/TypeScript dashboard over FastAPI SSE, integrating with a PostgreSQL database to surface per-LLM-call token usage and derived cost in real time via OpenTelemetry GenAI spans exported to Dynatrace.

Experience

Stevens Institute of Technology | **internal-notes-generator** **May 2025 – May 2026**

Technology Support *Hoboken, NJ, USA*

- Cut a multi-hour weekly scheduling-notes workflow down to minutes by building a full-stack automation tool (Next.js) that parsed Coursedog CSV exports across 15+ scheduling-logic cases and submitted formatted event notes via authenticated REST APIs, backed by unit tests and peer code review.
- Reduced environment setup time ∼80% by writing Python and PowerShell scripts that automated directory, tool, and browser-session launches across 10+ preconfigured environments spanning Windows and Linux, including network drive mapping and shared storage provisioning.

IGSCF | **IGSCF_website** **Feb 2026 – April 2026**

Full-Stack Developer (Freelance) *Hoboken, NJ, USA*

- Led end-to-end customer engagement as the sole technical point of contact for a nonprofit client, running requirements sessions and live demos and translating non-technical stakeholder needs into a shipped production application.
- Owned the authentication and authorization architecture end-to-end on a cloud platform: Firebase Auth with role-based access control and Firestore security rules enforcing public reads and authenticated writes, powering a 5-content-type admin CRUD portal.

Education

Stevens Institute of Technology **May 2026**

Master of Science, Computer Science — GPA: 3.5/4.0 *Hoboken, NJ, USA*

St. Francis Institute of Technology **Jun 2024**

Bachelor of Engineering, Information Technology — GPA: 7.98/10 *Mumbai, MH, India*

Technical Skills

Generative AI: OpenAI, Claude Code, RAG, Vector Search (Chroma, Azure AI Search, pgvector), Embeddings, Multi-Agent Systems, LangGraph, Pydantic, Prompt Engineering
Programming Languages: Python, JavaScript, TypeScript, SQL, Bash
Frameworks & Libraries: FastAPI, React.js, Next.js, REST APIs
Cloud & DevOps: AWS (ECS, Fargate, IAM, Secrets Manager), Azure, Terraform, Docker, GitHub Actions (CI/CD)
Observability & Testing: Datadog (LLM Observability, APM), OpenTelemetry, pytest
Databases: PostgreSQL, DynamoDB, Redis
Tools & Practices: Git, Linux/SSH, Agile (Scrum, Kanban), Code Reviews

Certifications

AWS Certified Solutions Architect – Associate (SAA-C03)